

Evaluating and Equivalent ExpressionsQuestion Bank - Additional 3.06 **ANSWER KEY***Show your work. Circle your final answer.***For questions 1 through 5, apply the laws of exponents to evaluate. Show your steps.**

1. $\frac{4^{-3}}{4^{-5}} \cdot 4^4$

$$4^2 \cdot 4^4 = 4^6 = 4,096$$

2. $\frac{((-5)^3)^4}{(-5)^2 \cdot (-5)^{-3}}$

$$\frac{(-5)^{12}}{(-5)^{-1}} = (-5)^{13} \quad (= -1,220,703,125)$$

3. $\left(\left(\frac{2}{3}\right)^0 \cdot \left(\frac{3}{2}\right)^{-2}\right)^3$

$$\left(\left(\frac{2}{3}\right)^2\right)^3 = \left(\frac{2}{3}\right)^6 = \frac{64}{729}$$

4. $\frac{\left(\frac{1}{3}\right)^2 \cdot \left(\frac{1}{3}\right)^3}{\left(\frac{1}{3}\right)^{-4}}$

$$\left(\frac{1}{3}\right)^9 = \frac{1}{19,683}$$

5. $\left(\frac{2^3 \cdot 2^4}{2^{-3}}\right)^{-2}$

$$(2^{10})^{-2} = 2^{-20} = \left(\frac{1}{2}\right)^{20}$$

Evaluate each expression in questions 6 through 9. Show all the steps.

6. $\frac{\left(\frac{1}{3}\right)^{-5} \cdot (3^7)^0}{(3^2)^{-3}}$

$$\frac{3^5 \cdot 1}{3^{-6}} = 3^{11} = 177,147$$

7. $\frac{2 \cdot 3^{-4} \cdot 2 \cdot 3^0}{(2 \cdot 3^2)^{-2}}$

$$\frac{2 \cdot 3^{-4}}{2 \cdot 3^{-4}} = 1$$

$$8. \quad \frac{\left[\left(\frac{1}{4}\right)^{-5} \cdot \left(\frac{1}{4}\right)^2\right]^2}{\left(\frac{1}{4}\right)^{-4}}$$

$$\frac{\left(\frac{1}{4}\right)^{-6}}{\left(\frac{1}{4}\right)^{-4}} = \left(\frac{1}{4}\right)^{-2} = 16$$

$$9. \quad \frac{(4^{-1})^3 \cdot \left(\frac{1}{4}\right)^{-4}}{4^3}$$

$$\frac{4^{-3} \cdot 4^4}{4^3} = 4^{-2} = \frac{1}{16}$$

Common bases.

10. Write each expression as a single power with the base given.

a. 81^3 with base 3

$$3^{12}$$

b. 64^3 with base 2

$$2^{18}$$

c. $4^2 \cdot 64^3$ with base 4

$$4^{11}$$

$$64 = 4^3.$$

d. $27 \cdot 3^{-5}$ with base 3

$$3^{-2}$$

11. Paul says $\left(\frac{1}{8}\right)^2 \cdot 2^{-2}$ and $\left(\frac{1}{4}\right)^4 \cdot 4^{-1}$ are equivalent because both can be written as 2^{-8} .

a. Write the first expression as a single power of 2. Show your work.

$$2^{-6} \cdot 2^{-2} = 2^{-8}$$

b. Write the second expression as a single power of 2. Show your work.

$$2^{-8} \cdot 2^{-2} = 2^{-10}$$

c. Is Paul correct?

No — only the first is 2^{-8} ; the second is 2^{-10} .

12. Which expression is equivalent to $\frac{3^5}{6^2}$?

- ☒ A. $\frac{3^3}{2^2}$
- ☐ B. $\left(\frac{3}{6}\right)^3$
- ☐ C. 3^3
- ☐ D. $\frac{3^7}{2^2}$

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$$6^2 = 3^2 \cdot 2^2, \text{ so } 3^5/(3^2 \cdot 2^2) = 3^3/2^2.$$

13. Find the value of n that makes each statement true.

a. $27^n = 3^{12}$

$n = 4$

b. $\left(\frac{1}{4}\right)^n = 2^{10}$

$n = -5$